The Hong Kong Polytechnic University

Department of Electronic and Information Engineering

EIE2111 Assignment 1: Address Book

(Due date to submit: Check the course information)

Introduction

A software-based address book can store personal contact information such as name, telephone number and e-mail address of individuals. Information can be retrieved by searching the address book content based on the input name, telephone number or e-mail address of the person of interest.

Programming Description

Section A: Class (20 marks)

Design and implement a simple `Record` class for static library with the following requirements:

1. Create a new project called `Record`.

2. Use `Record` as the class name for the library, and create a header file called `Record.h` and a class file called `Record.cpp` file to implement the `Record` class.

3. For private members of the class, it should include:
   (a) A text string to store the name of a person.
   (b) Another text string to store the e-mail address (e.g., abc@def.com) of the person.
   (c) An integer variable to store the 8-digit telephone number of the person.

4. For public members of the class, it should include:
   (a) A pair of `set` & `get` functions to operate on the text string for name.
   (b) A pair of `set` & `get` functions to operate on the text string for e-mail.
   (c) A pair of `set` & `get` functions to operate on the integer variable for 8-digit phone number.

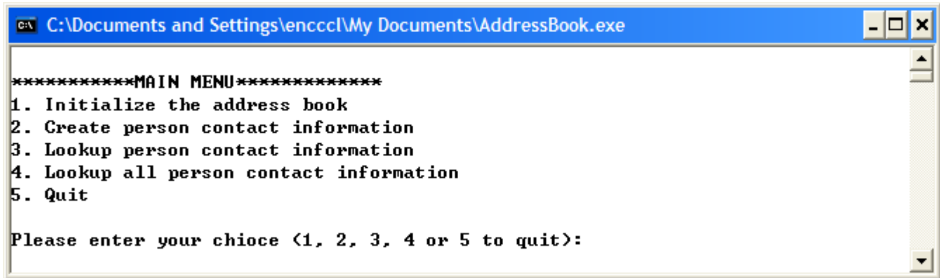
5. Build the `Record` class into a static library (e.g. `Record.lib`).

Section B: Main Program (70 marks)

By using Microsoft Visual Studio 2019, design and implement a simple program with the following requirements:

1. Create a new project called `AddressBook`. The project should be an empty project.

2. At the beginning, the project should show the following main menu on the screen:



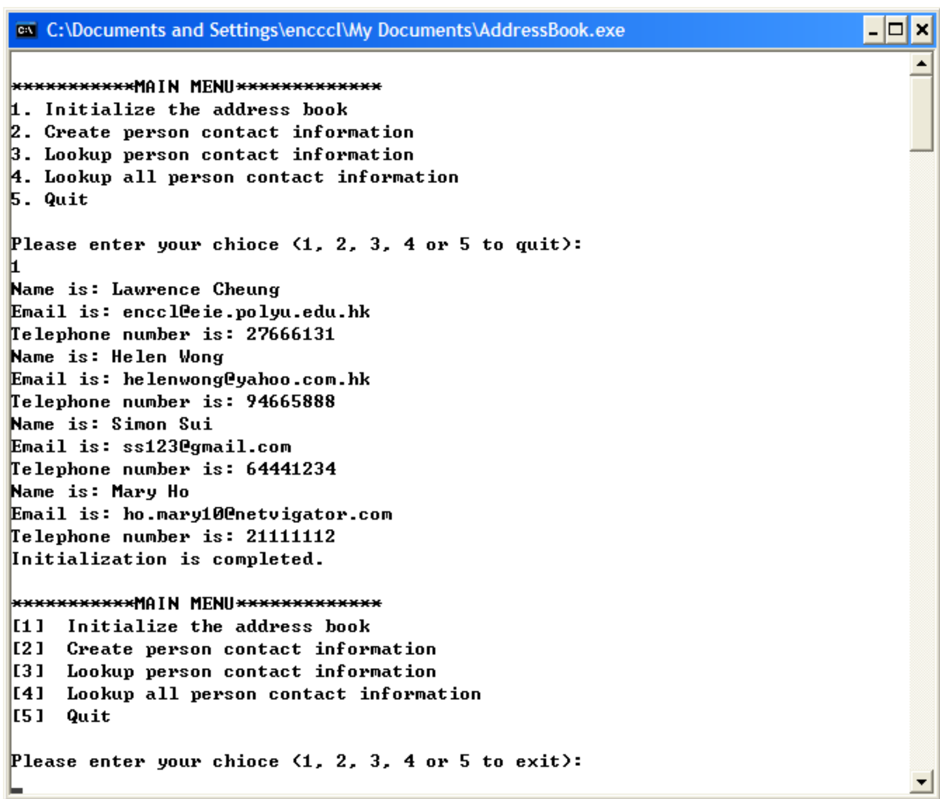
```
C:\Documents and Settings\enccl\My Documents\AddressBook.exe

***********MAIN MENU*************
1. Initialize the address book
2. Create person contact information
3. Lookup person contact information
4. Lookup all person contact information
5. Quit

Please enter your chioce (1, 2, 3, 4 or 5 to quit):
```

3. In this main menu, we have the following requirements: The end user is expected to enter an integer of value 1, 2, 3, 4 or 5 only. When the user enters an input other than that, an error message “Invalid input. Please enter again!” should be displayed on the screen and then the main menu is ***displayed again***. (10 marks)

4. If the choice is **1**, the following records should be added to the address book:



```
C:\Documents and Settings\enccl\My Documents\AddressBook.exe

***********MAIN MENU*************
1. Initialize the address book
2. Create person contact information
3. Lookup person contact information
4. Lookup all person contact information
5. Quit

Please enter your chioce (1, 2, 3, 4 or 5 to quit):
1
Name is: Lawrence Cheung
Email is: enccl@eie.polyu.edu.hk
Telephone number is: 27666131
Name is: Helen Wong
Email is: helenwong@yahoo.com.hk
Telephone number is: 94665888
Name is: Simon Sui
Email is: ss123@gmail.com
Telephone number is: 64441234
Name is: Mary Ho
Email is: ho.mary10@netvigator.com
Telephone number is: 21111112
Initialization is completed.

***********MAIN MENU*************
[1]  Initialize the address book
[2]  Create person contact information
[3]  Lookup person contact information
[4]  Lookup all person contact information
[5]  Quit

Please enter your chioce (1, 2, 3, 4 or 5 to exit):
```

After initializing the address book, all records should be displayed, a message “Initialization is completed.” should be displayed on the screen and then the main menu is ***displayed again***. To store new records, you should create a 100-element array of objects at the beginning. Note that the above records should be found in the array of objects. (5 marks)

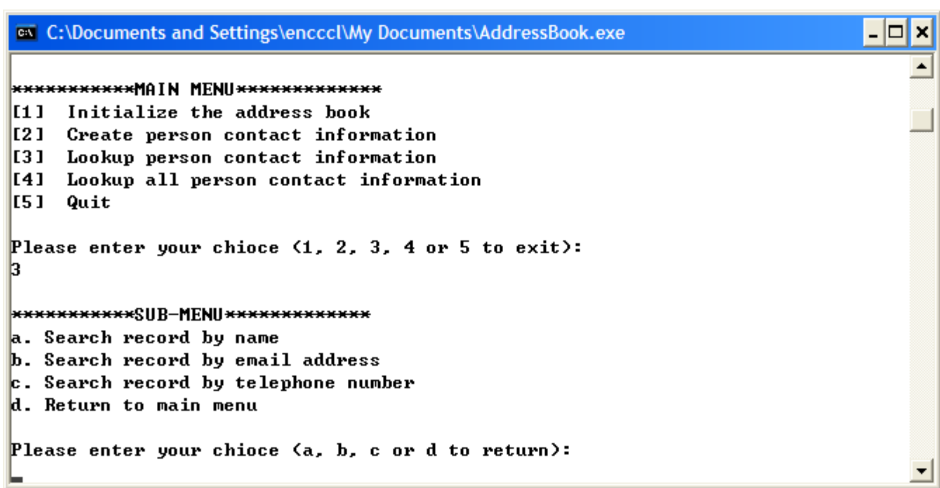
5. If the choice is **2**, the program should display a message "Please enter the total number of records to be created", and then store the input integer. You may set the maximum number of new records is 10. If the input is invalid (e.g., not an integer) or an integer greater than 10, an error message "The input is invalid. It must be an integer not greater than 10." should be displayed on the screen and then ask for the input again. After that, a ***sub-menu*** should be displayed for the end user to input full name(s), email address(es), and 8-digit telephone number(s) correspondingly. Note that input validation must be processed. Note that you are not required to check whether the remaining spaces of the array are large enough to store new records.

   - For a full name, only space characters and letters (i.e., 'a' to 'z', 'A' to 'Z') are allowed. Other characters are invalid.
   - For an email address, one character '@' must be found. Moreover, there must be some characters before and after the character '@' (e.g., abc@def.com).
   - For a telephone number, it must be an 8-digit integer and the first digit cannot be zero.

   Finally, the program should display a message "All new records are saved!" on the screen and return to the ***main menu***. (20 marks)

   You may find the following information is useful: the i^{th} character of a string `TestString` is `TestString.at(i)`.

6. If the choice is **3**, a ***sub-menu*** should be displayed:



   The end user is expected to enter a character of value a, b, c or d only. When the user enters an input other than that, an error message "Invalid input. Please enter again!" should be displayed on the screen and then the sub-menu is ***displayed again***.

   - If the choice is **a**, the program should display a message "Please enter a name to search" on the screen and then store the input name. After that, the program should search the array of records to find a match. If no match is found, the program should display a message "Sorry, no record is found!" on the screen and return to the ***sub-menu***. If a match is found, the program should display a message “A record is found!” on the screen and the information of the record. Finally, the program should return to the ***sub-menu***.

- If the choice is **b**, the program should display a message “Please enter an email address to search” on the screen and then store the input email address. After that, the program should search the array of records to find a match. If no match is found, the program should display a message “Sorry, no record is found!” on the screen and return to the ***sub-menu***. If a match is found, the program should display a message “A record is found!” on the screen and the information of the record. Finally, the program should return to the ***sub-menu***.

- If the choice is **c**, the program should display a message “Please enter a telephone number to search” on the screen and then store the input telephone number. After that, the program should search the array of records to find a match. If no match is found, the program should display a message “Sorry, no record is found!” on the screen and return to the ***sub-menu***. If a match is found, the program should display a message “A record is found!” on the screen and the information of the record. Finally, the program should return to the ***sub-menu***.

- If the choice is **d**, the program should return to the ***main menu***. (25 marks)

7. If the choice is **4**, the program should display the information of all records on the screen and then return to the ***main menu***. (5 marks)

8. If the choice is **5**, a message “Goodbye!” should be displayed on the screen and the program exits (i.e. return back to the command prompt). (5 marks)

Section C: The program Flow (10 marks)

Use Microsoft Word or PowerPoint to show the pseudo code or UML activity diagram of the program flow. The design of the program structure is important when assessing your work. It is inappropriate to write all your programs within a single `main()` function.

Instructions

Zip your C++ projects (Section A and B) and the program flow file into one single file. Submit the zip file to Blackboard on or before the deadline. The deadline can be found in the course information.

Appendix: FAQ

Q: I find that I get the wrong result if I use the function `getline` more than once. Why?

A: I find the function works very well. Here is my small testing program and it may be helpful for you.

```
#include <iostream>
using std::cout;
using std::cin;
using std::endl;

#include <string>
using std::string;
using std::getline;

int main()
{
    string name1, name2;

    cout << "Enter the first string: ";
    getline(cin, name1);
    cout << "Enter the second string: ";
    getline(cin, name2);
    cout << "result = " << name1 << " and " << name2 << endl;
    return 0;
}
```

I know you may have a problem if you use both `cin` and `getline` to read the input. If you use `cin` first and then `getline`, you need to put a dummy `getline` between them to remove the newline character from `cin`.

Q: Is Section A related to Section B?

A: In your assignment, you are asked to write a program as a software-based address book. To write the program, you need to create a class for data records (Section A) and then use the class to create a 100-element array for your program (Section B). So you should finish Section A first and then Section B.

Lawrence Cheung
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